Try-catch & Firebase authentication

Firebase Development

NIFR 2023 - https://github.com/nicklasdean/firebase

Agenda

Error handling & Firebase auth

- Error handling in programming
- Try/Catch/Finally in javascript
- Authentication
- Firebase Auth

What are errors in programming?

Error handling

- Before running
  - Syntax errors
  - Compilation errors
  - Bundling error
- Runtime / While running
  - Ressource not available
  - Wrong input
  - TypeError

```
  168:36  error  Parsing error: Unexpected token .

✖ 1 problem (1 error, 0 warnings)

File: src/templates/post.js

 ERROR #98123  WEBPACK

Generating development JavaScript bundle failed
```

```
Uncaught TypeError: guessSubmit.addeventListener is not a function      number-game-errors.html:86:15
    <anonymous>   …ion-to-js-1/troubleshooting/number-game-errors.html:86
  [Learn More]
```

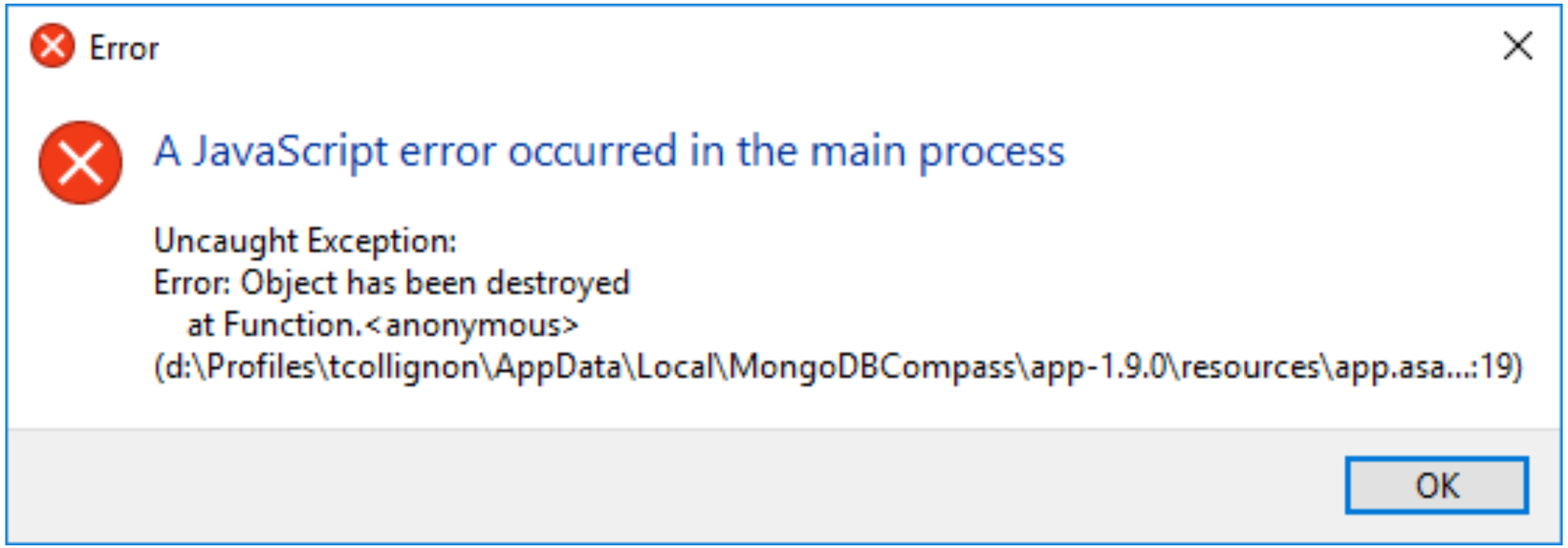
```
> var testArray=null;
  if(testArray.length===0){
     console.log("Array is empty");
  }
✖ ▼Error
     line: 3
     message: "'null' is not an object (evaluating 'testArray.length')"
     sourceId: 2056192896
   ▶ __proto__: Error
>
```

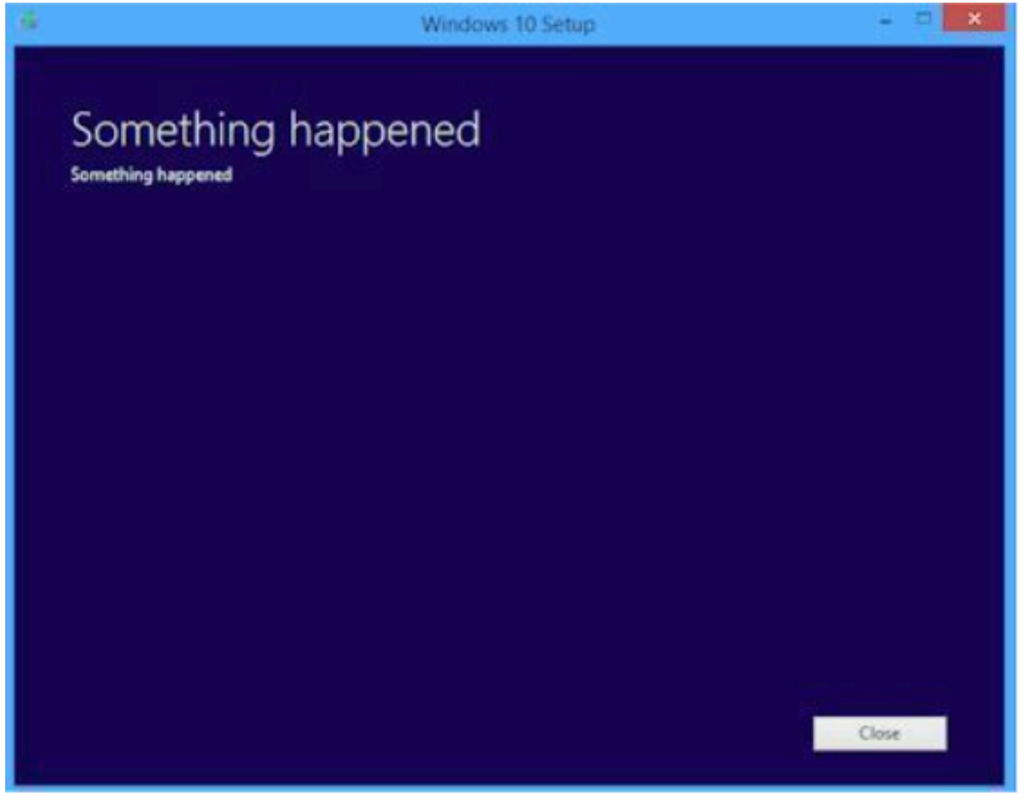
Some errors will stop execution

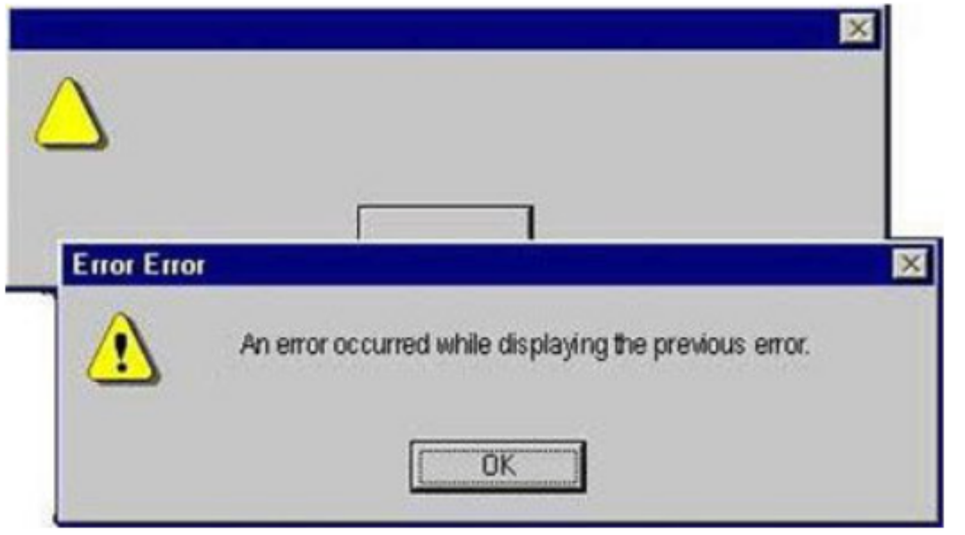
Examples

What to do with runtime errors?

Informing the user




Windows 10 Setup
Something happened
Something happened
Close


Error Error
An error occurred while displaying the previous error.
OK

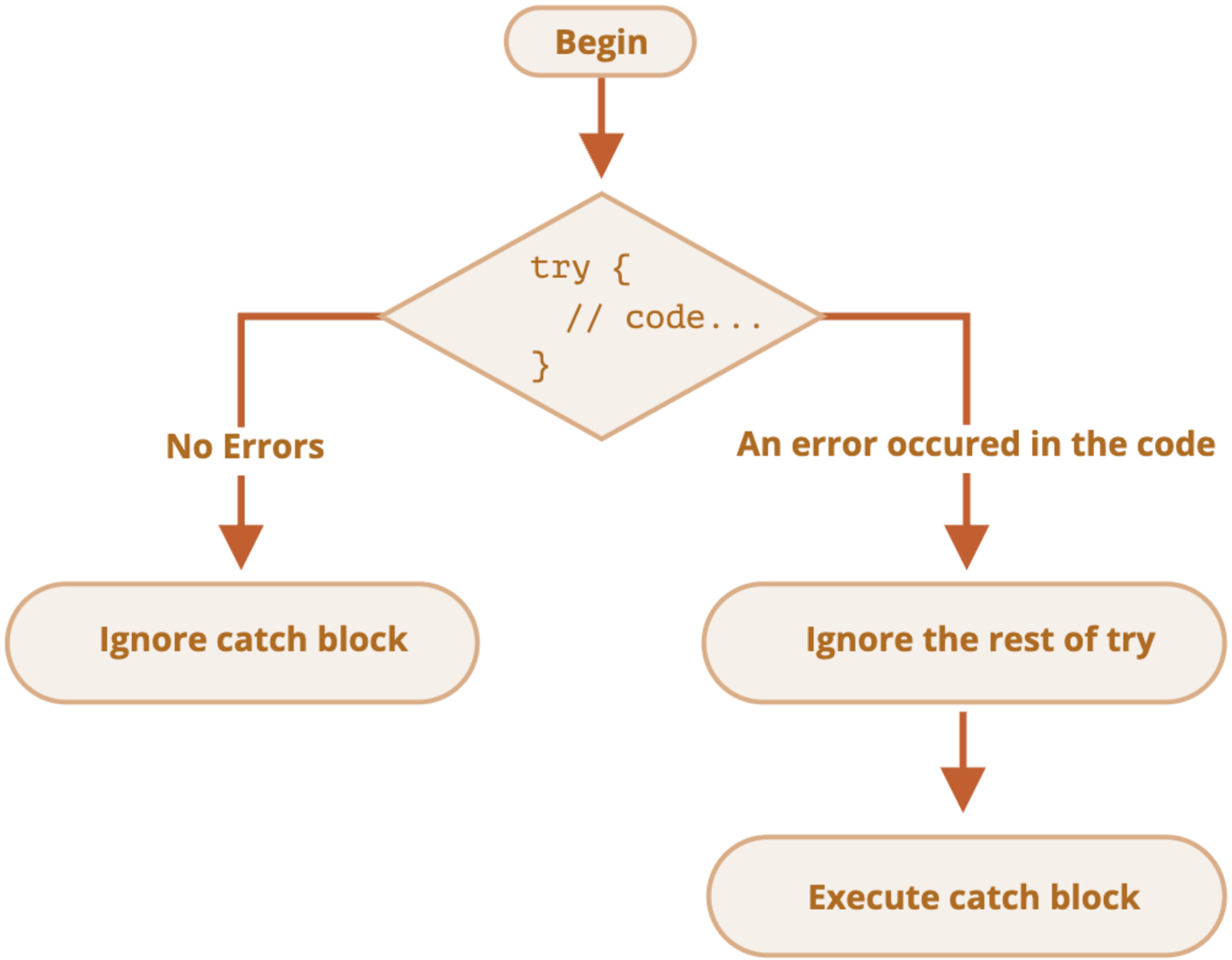
Windows XP
Task failed successfully.
OK

Alternative?

Handling the error gracefully

```
try{
    //Code that fetches data from API
}
catch(err){
    //Code that fetches same data from other API
}
```

```
try{
    //Code
}
catch(err){
    //Code
}
```



Web modular API | Web namespaced API | Swift | Objective-C | Kotlin+KTX Android | Java Android | C++ | More

Learn more about the tree-shakeable modular Web API and upgrade from the namespaced API.

```javascript
import { collection, addDoc } from "firebase/firestore";

try {
  const docRef = await addDoc(collection(db, "users"), {
    first: "Ada",
    last: "Lovelace",
    born: 1815
  });
  console.log("Document written with ID: ", docRef.id);
} catch (e) {
  console.error("Error adding document: ", e);
}
```

add_ada_lovelace.js

https://firebase.google.com/docs/firestore/quickstart#add_data

Try/Catch Exercises

Firebase authentication

BAAS auth

- E-mail authentication module in Firebase
- Part of the Firebase suite
  - Authentication “out of the box”
- Pre-requisites:
  - Webpack/NPM/Firebase

Web modular API Web namespaced API

Learn more about the tree-shakeable modular Web API and upgrade from the namespaced API.

```
import { getAuth, createUserWithEmailAndPassword } from "firebase/auth";

const auth = getAuth();
createUserWithEmailAndPassword(auth, email, password)
  .then((userCredential) => {
    // Signed up
    const user = userCredential.user;
    // ...
  })
  .catch((error) => {
    const errorCode = error.code;
    const errorMessage = error.message;
    // ..
  });
```

auth_signup_password.js

https://firebase.google.com/docs/auth/web/start

Web modular API Web namespaced API

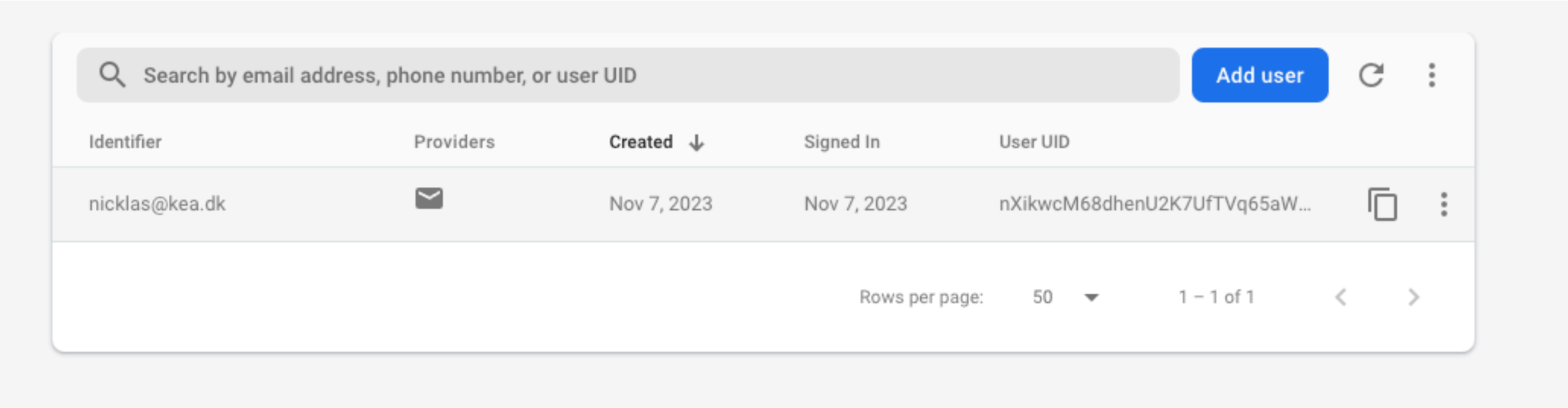
Learn more about the tree-shakeable modular Web API and upgrade from the namespaced API.

```
import { getAuth, signInWithEmailAndPassword } from "firebase/auth";

const auth = getAuth();
signInWithEmailAndPassword(auth, email, password)
  .then((userCredential) => {
    // Signed in
    const user = userCredential.user;
    // ...
  })
  .catch((error) => {
    const errorCode = error.code;
    const errorMessage = error.message;
  });
```

auth_signin_password.js

https://firebase.google.com/docs/auth/web/start

Cloud Firestore data model example

```
note{
    "user-id":"nXikwcM68dhenU2K7UfTVq65aW82",
    "note-title":"great idea!"
    "note-body":"Let the cat out of the bag"
}
```